

An Empirical Specification for the V -Extension

Measuring Per-Capita Structural Novelty to Test Core Claim 6
(the companion instrument to the WS3 C - V primer)

Originality Program — Whitespace 2 $\rightarrow$ Whitespace 3 bridge

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1 Purpose, placement, and the one-sentence claim

The WS3 primer (cv-reconciliation-primer.tex) defines two output gauges, cumulative complexity $C(t)$ and per-capita variance generation $V(t)$, and shows that Whitespace 2 (WS2) pinned the collective side (C -family breadth W , canonical concentration H , demographic diversity D ; all rising) but left the per-capita gauge V *unmeasured*. **Core Claim 6** of that primer is therefore a prediction rather than a result:

per-capita *structural* novelty $V^{\text{struct}}(t)$ is declining—strongest where the canon is most concentrated (Physics)—while per-capita *topical* novelty $V^{\text{lat}}(t)$ need not decline.

This document specifies the empirical instrument that converts Core Claim 6 into a falsifiable test, reusing data already in hand.

Placement. Operationally this is a **WS2 Phase 2.4** (a per-capita extension of the Phase 2.2/2.3 machinery); scientifically it is the **empirical anchor for WS3**. It requires *no new embedding*: the base-1M SciNCL, Qwen3, and SPECTER2 vectors, the 1M source corpus (references + concepts + authorships), the Phase-1.3 demographic substrate, and the reference-canoncity machinery all already exist. Marginal compute is a citation-graph inversion and panel regressions—\$0 (one optional robustness pull excepted, §9).

The one design principle. Everything below is built to respect the same distinction the theory turns on (primer Remark 2.3): an author’s output has a *content* channel (what they work on $\rightarrow V^{\text{lat}}$) and an *attachment* channel (what they build on / cite $\rightarrow V^{\text{struct}}$). The instrument must measure the two *separately* and per capita, and—per the hard-won WS2 cautions—must measure the attachment channel in at least one *embedding-free* way, because WS2 showed embedding-based semantic trends are model-dependent.

2 Estimands: from model quantities to observables

Fix a study window (1970–2023, trimming the incomplete final year). Index papers p , authors a , years t ; let T_p be paper p ’s team size (author count) and $F(p)$ its field/subfield.

Estimand 1 (Per-capita structural novelty). For author a in year-window t , $V_a^{\text{struct}}(t)$ is the mean over a ’s papers in t of *persisting attachment-deviant novelty*: a paper contributes when it (i) builds on non-canonical antecedents (low empirical canon-alignment $\hat{\gamma}$, §4.2) and (ii) that move *persists* (§4.3). Its field-year aggregate $V^{\text{struct}}(t)$ is the per-capita mean over authors active in (F, t) . The estimand of interest is the **trend** dV^{struct}/dt and its **gradient in canon concentration** H .

Estimand 2 (Per-capita topical novelty — the contrast). $V_a^{\text{lat}}(t)$ is the analogous quantity built from *content* novelty (distance from topical precedent, §4.1) rather than attachment deviance. Its trend is the **control contrast**: Core Claim 6 predicts V^{struct} falls while V^{lat} does not.

Remark 2.1 (Why the two must be measured with the right lens). WS2 established that *topical* embeddings (SciNCL, Qwen3) track the content channel and a *citation-tuned* embedding (SPECTER2) tracks the attachment channel. Accordingly V^{lat} is measured on SciNCL/Qwen3 and V^{struct} on SPECTER2 and on embedding-free reference statistics. Using the wrong lens would re-run the exact confusion WS2 diagnosed.

Artifact	Contents / role
base-1M vectors (SciNCL, Qwen3, SPECTER2)	$902,531 \times 768$, row-aligned to <code>metadata.parquet</code> ; SciNCL/Qwen3 $\rightarrow V^{\text{lat}}$, SPECTER2 $\rightarrow V^{\text{struct}}$ (embedding arm)
1M source corpus	<code>referenced_works_json</code> (attachment / citations), <code>concepts_json</code> (topics, subfields), <code>authorships</code>
demographic substrate (Phase 1.3)	per-author <code>author_id</code> , <code>min_year</code> (career stage), <code>primary_country</code> , <code>gender</code> ; ORCID flags for a well-disambiguated subset
canon machinery	<code>reference_canonicity</code> : per-year reference-target frequencies, Gini H , top- N set $\rightarrow$ the canon $K(t)$ for $\hat{y}$
K=50 clusters + subfield maps (Phase 2.3)	data-driven and concept-level subfields for the H -gradient test
in-sample citation graph	invert <code>referenced_works_json</code> across the corpus $\rightarrow$ time-resolved forward uptake (the persistence filter)

Table 1. Inputs. Nothing here requires new embedding or new pulls; the only derived object is the in-sample forward-citation graph (an edge inversion).

3 Data inventory (all existing)

4 Measurement

4.1 Topical novelty $\rightarrow V^{\text{lat}}$

Measure 1 (Distance-to-precedent (content)). For paper p published in year t_p , field F , with topical embedding x_p (SciNCL; Qwen3 as robustness), define

$$n^{\text{lat}}(p) = 1 - \max_{q \in \mathcal{P}(p)} \cos(x_p, x_q), \quad \mathcal{P}(p) = \{q : t_q < t_p, F(q) = F\} \text{ (precedent set)}, \quad (1)$$

optionally averaged over the κ -nearest precedents rather than the single max (robustness). Higher = more topically novel relative to prior work. For tractability $\mathcal{P}(p)$ is a seeded random sample (fixed size) of prior in-field papers; report the sample size.

4.2 Structural novelty $\rightarrow V^{\text{struct}}$ (three operationalizations)

The attachment channel is measured *three* ways—two embedding-free, one embedding—so the headline cannot hinge on any single lens (the WS2 caution).

Measure 2 (Off-canon reference share — the empirical $1 - \gamma$). Let $R(p)$ be p ’s reference set and $K(t_p)$ the canon as of t_p (top-weighted works using *only* data up to t_p ; §6 on leakage). The empirical canon-alignment is $\hat{\gamma}(p) = |R(p) \cap K(t_p)| / |R(p)|$ (the estimand of primer Eq. 6), and

$$n_{\text{oc}}^{\text{struct}}(p) = 1 - \hat{\gamma}(p) \in [0, 1]. \quad (2)$$

This is the most direct image of the model’s structural-deviance quantity: how much of p ’s foundation lies off the shared canon.

Measure 3 (Reference atypicality — novel recombination (Uzzi-style)). For each unordered pair of references $\{u, v\} \subseteq R(p)$, compute the z -score of their observed co-reference frequency against a degree-preserving null (co-reference expected from each work’s citation frequency). Summaries per paper: the *median* z (conventionality) and the *10th-percentile* z (the atypical tail). Structural novelty $n_{\text{atyp}}^{\text{struct}}(p)$ is decreasing in median z / increasing in a left (atypical) tail. Embedding-free; captures *recombination* novelty the off-canon

share misses.

Measure 4 (Citation-geometry distance (SPECTER2)). With SPECTER2 (citation-tuned) embedding y_p , define $n_{\text{emb}}^{\text{struct}}(p) = 1 - \max_{q \in \mathcal{P}(p)} \cos(y_p, y_q)$ exactly as in Measure 1 but on the attachment-tuned family. This is the embedding arm; its agreement or disagreement with the two reference-based measures is itself a headline diagnostic.

4.3 The persistence filter — the “ $\geq k$ generations” gate

A novelty counts only if it *sticks* (primer Estimand 1); a one-off attachment nobody adopts is churn, not V .

Measure 5 (Uptake-within-window persistence). Using the in-sample forward-citation graph, let $u_w(p)$ be p ’s forward citations accrued within a fixed w -year post-publication window (e.g. $w = 5$), *age-restricted* so every paper has a full window (drop papers whose window extends past the snapshot). Define the persistence weight $s(p) = \mathbf{1}[u_w(p) \geq \theta_w]$ or a smooth $s(p) = u_w(p)/(u_w(p) + \bar{u}_w)$, where thresholds/scales are computed *within* (field, publication-year) cells to remove field/era base-rate differences.

Remark 4.1 (Persistence, not the CD-index). Per WS3’s own pre-commitment (“frame V around any novelty metric, not specifically CD-index”), persistence is a robust forward-uptake filter, not a disruption index. A CD/disruption variant may be reported as a *secondary* column with the Petersen–Holst caveats attached, never as the headline.

4.4 Per-capita aggregation and team size

Measure 6 (Per-paper persisting novelty, and per-capita roll-up). Per-paper persisting structural novelty is $v^{\text{struct}}(p) = n^{\text{struct}}(p) \cdot s(p)$ (analogously for lateral). Two per-capita estimands:

- (i) **Paper-level with team-size control** (primary): analyze $v^{\text{struct}}(p)$ with $\log T_p$ as a covariate.
- (ii) **Fractional per-author** (robustness): attribute $v^{\text{struct}}(p)/T_p$ to each author, then average within (a, t) to form the author panel $V_a^{\text{struct}}(t)$.

Because mean team size rose sharply over the window—and larger teams are less disruptive per capita (WWE)—team size is both a *confound to control* and *part of the mechanism*; Hypothesis 4 tests whether the decline survives *within* team-size strata.

5 Estimator and pre-registered hypotheses

The headline estimator is a panel regression of per-paper (or per-author) persisting novelty on year, with fixed effects and controls:

$$v^{\text{struct}}(p) = \beta_0 + \beta_1 t_p + \beta_2 \log T_p + \beta_3 \log(\text{field volume}_{F, t_p}) + \alpha_{a(p)} + \delta_F + \varepsilon_p, \quad (3)$$

with author fixed effects α_a (within-author identification) and field effects δ_F . Inference by **permutation** (shuffle t_p within field) plus the **absolute** change over the window; **VIF reported** for $(t, \log T, \log \text{volume})$; every headline run repeated for ≥ 2 **structural measures** (Measures 2 to 4).

Hypothesis 1 (Primary: structural novelty declines). $\beta_1 < 0$ for v^{struct} , *robust across (a) the off-canon share and the atypicality measure (embedding-free) and the SPECTER2 measure, and (b) permutation significance with a non-negligible absolute effect.*

Hypothesis 2 (Contrast: topical novelty does not decline). $\beta_1 \geq 0$ or non-significant for v^{lat} (SciNCL and Qwen3). *The dimension-specificity ($V^{\text{struct}} \downarrow, V^{\text{lat}}$ spared) is the signature that distinguishes the mechanism*

from a generic novelty decline.

Hypothesis 3 (Differential in canon concentration). *Interacting year with subfield canon concentration H_F (from Phase 2.3), the year $\times$ H coefficient is negative: the structural decline is steeper where the canon is more concentrated (Physics > CS), matching the WS2 aggregate finding.*

Hypothesis 4 (Within-team-size robustness). $\beta_1 < 0$ survives within team-size strata (and with the fractional per-author estimand), so the decline is not merely mechanical team-size growth.

Hypothesis 5 (Mechanism link to κ). *Within author (and within team-size), off-canon reference share falls as the subfield canon concentrates: a direct estimate of the model’s conformity acting on the attachment channel (κ suppressing low- γ output).*

Negative controls (pre-registered). (i) Year-shuffle placebo: the trend must vanish. (ii) Random-canon placebo: recompute $\hat{\gamma}$ against a size-matched *random* reference set in place of $K(t)$; the structural trend must attenuate to null (confirming it is the *canon* that bites, not reference growth). (iii) A pure-noise novelty series must show no trend.

6 Confounds and controls

Each control carries a specific WS2/Phase lesson.

- C1. Citation accrual / age.** Forward-uptake persistence is age-confounded (recent papers have less time). Use *fixed w-year windows*, age-restrict to papers with a complete window, and trim the incomplete final years—the Phase-1.4/2.2 age-restriction lesson (raw `cited_by_count` is invalid across cohorts).
- C2. Ratio $\neq$ control.** Report *absolute* per-capita trends for V^{struct} and V^{lat} *separately*; never divide one novelty channel by another (the Phase-2.2 denominator-confound lesson).
- C3. ≥ 2 (here 3) structural measures.** Off-canon share + atypicality (embedding-free) + SPECTER2 (embedding). A decline seen only in the embedding arm is model-specific, not a finding (the SPECTER2/Qwen3 sign-reversal lesson).
- C4. Ill-conditioned σ .** Judge trends by permutation significance *and* absolute magnitude, never a standardized effect on a near-flat series (the Phase-2.3 ill-conditioned- σ artifact).
- C5. Collinearity.** Year, log team size, and log volume all trend together; report VIF and prefer within-stratum comparisons where $\text{VIF} \gtrsim 10$ (the WS-F unidentifiability lesson).
- C6. Canon leakage / endogeneity.** $K(t_p)$ and the atypicality null use *only* data up to t_p (no look-ahead); the canon a paper is scored against is the canon it faced.
- C7. Author disambiguation.** Author FEs and career panels are sensitive to OpenAlex split/merge error ($\sim 3\%$ cross-era mergers, Phase 1.3). Repeat on the **ORCID-backed, career-length-screened** subset as robustness.
- C8. Survivorship / selection.** Long-observed authors are selected; use within-author FEs, entry-cohort controls, and report the estimate on *entrants* separately from incumbents.
- C9. Field composition.** CS/Physics mix and subfield entry shift over time; field (and subfield) FEs, plus the per-field decomposition.

Axis	Settings
structural measure	off-canon share / atypicality / SPECTER2 (all three)
topical measure	SciNCL / Qwen3
subfield definition	concept level-1 / K=50 clusters (Phase 2.3 pair)
persistence window w	{3, 5, 10} years (sweep)
per-capita form	paper-level + log T control / fractional per-author
disambiguation	full / ORCID-backed subset
field	pooled (field FE) / CS-only / Physics-only
canon depth α	top- $\{1, 5, 10\}$ % for $K(t)$

Table 2. The pre-registered robustness grid. The headline (Hypothesis 1) must hold across the structural-measure and disambiguation axes at minimum; the gradient (Hypothesis 3) is read on the field / subfield axes.

7 Robustness grid

8 Pre-registered evaluation gates

Gate 1 (Instrument validity). *The three structural measures are pairwise positively correlated at the paper level (they measure a common construct), and the topical measures track the Phase-2.2 aggregate W trend on a re-aggregation check.*

Gate 2 (Negative controls pass). *Year-shuffle and random-canon placebos return null; the random-canon placebo specifically kills the structural trend (isolating the canon as the operative object).*

Gate 3 (Headline). *Hypothesis 1 holds (structural decline, ≥ 2 measures, permutation-sig + material magnitude) and Hypothesis 2 holds (topical spared). Together these confirm Core Claim 6; either failing is reported honestly.*

Gate 4 (Mechanism + gradient). *Hypotheses 3 to 5 adjudicate whether the decline is canon-driven, per-capita-real (not team-size-mechanical), and localized in H .*

Escape hatches (pre-registered). If author disambiguation is too noisy for author FEs even on the ORCID subset, fall back to *team-level* (paper) panels with cohort controls (diagnose, don’t force the author panel—the Phase-1.3 “diagnose before escalating” lesson). If the in-sample forward graph is too sparse for a w -year filter in early decades, widen w or restrict the persistence-filtered analysis to the post-1985 window and report the unfiltered novelty trend for the full window alongside.

9 Feasibility, cost, and the one data decision

All measures derive from existing artifacts (Table 1); the only new object is the in-sample forward-citation graph (invert `referenced_works_json` over the corpus—an $O(\text{edges})$ join, laptop-scale, \$0). Panel regressions are trivial. **The one decision:** the persistence filter’s forward citations. *Primary (pre-registered):* the *in-sample* graph—self-contained and \$0, but it undercounts uptake (only citations from papers inside the 1M sample) and its coverage may drift with sample density, so persistence thresholds are computed within (field, year) cells to absorb level shifts. *Optional robustness:* an OpenAlex forward-citation pull for complete, time-resolved uptake (a bounded API job) to confirm the in-sample trend is not a coverage artifact. Any bounded coverage caps are logged, never silently applied (the no-silent-truncation rule).

Outcome	Reading for WS3 (primer Core Claim 6 / Estimand 1)
$V^{\text{struct}} \downarrow$, V^{lat} spared, gradient in H , ≥ 2 measures agree	Core Claim 6 confirmed on data. The C – V reconciliation’s κ -on-attachment mechanism is empirically instantiated; WS3 leads with a measured, not merely modeled, V decline.
V^{struct} flat/up despite $H \uparrow$	Falsifies the CD-mechanism reading. Canon concentration does not suppress per-capita structural novelty; WS3 must down-weight the canon-deviation κ and privilege another mechanism (visible-consensus / attention).
decline only in the SPECTER2 arm	model-specific, <i>not</i> a finding (repeat of the WS2 embedding-sensitivity result); report as inconclusive.
decline vanishes within team-size strata	the “small teams disrupt” pattern is carried by team-size growth, not a within-stratum conformity effect — a distinct (and publishable) refinement of WWE’s per-capita story.

Table 3. How each outcome feeds back into the WS3 primer. The instrument is designed so that *every* outcome is informative for the theory—confirming, falsifying, or relocating the operative κ mechanism.

10 Interpretation matrix: reading outcomes back onto WS3

One-sentence summary. Measure, per author/team and per year, how far scientific output strays from the canonical way of building (*attachment*, three ways) versus from prior topics (*content*), filter for what *persists*, and test whether the former—not the latter—declines and does so most where the canon is strongest: the single measurement that turns the WS3 primer’s Core Claim 6 from a prediction into a result.